LLM-Distilled Semantic Knowledge for Task-Aware Object Selection

ABSTRACT

The conventional object detection frameworks identify the object in a scene but they are incapable of selecting the particular object for a particular task. This paper presents a training free framework for selecting an object based on the given input query using an offline 47KB distilled knowledge bank. The objects are detected using YOLOv8-nano detector and the scores are compared against the given task using cosine similarity between Sentence-BERT encoded embeddings. The fusion of semantic relevance scores and detection confidence produces a final ranked selection with a reference of a threshold score below which no objects will be selected. This approach is experimented across all 14 COCO-Tasks. The proposed method operates entirely on CPU.

INTRODUCTION:

Choosing an object based on the given task is an important challenge in a computer vision, as just identifying an object dose not reveal whether it us suitable for a given specific task. Sawatzky et al. used the method of GGNN, but their approach requires task-annotated training data and it is computationally expensive.

We proposed a training-free approach that uses LLM-generated semantic descriptions to build a compact knowledge bank. The YOLOv8-nano detects objects on a CPU, while sentence embeddings and cosine similarity and measure their suitability for a given task. The weighted score ranks the detected objects, with a threshold allowing the system to understand that no suitable object is present in the scene.

LITERATURE SURVEY:

[1] Sawatzky et al. – Task-Driven Object Detection
Introduced the COCO-Tasks dataset with ~40,000 images covering 14 manipulation and navigation tasks for task-appropriate object selection.
Used a Gated Graph Neural Network (GGNN) to model detected objects as graph nodes and propagate scene context through iterative message passing.
Achieved 32.6% mAP, outperforming classification baselines by 3.8 points, though detector class errors can introduce inference noise.

[3,4] Visual Question Answering and Scene Understanding
Early VQA methods combined CNN-based visual features with language representations to answer natural-language questions about images.
Bottom-up and top-down attention methods further improved scene understanding by combining object-level visual features with linguistic context.
Unlike VQA, task-driven object selection must produce a localised bounding box corresponding to the queried task.

[6–9] Object Affordance Detection
CNN-based methods detect affordance regions, while GGNN-based approaches model how object suitability for an action depends on surrounding objects and scene context.
Other approaches investigate functional understanding of tools using 3D object models and human demonstrations.
However, these methods are generally category-specific and have limited ability to generalise to arbitrary task queries at inference time.

[10] CLIP – Language-Vision Alignment
CLIP learns aligned visual and linguistic representations from large-scale image-text pairs and enables zero-shot visual classification using image-text similarity.
Task-aware extensions such as TaskCLIP apply similar alignment for task-based scenarios but require GPU inference.
Their high computational and power requirements (~150–200 W) make them unsuitable for resource-constrained mobile robotic platforms.

[11] Sentence-BERT – Language-Vision Alignment
Sentence-BERT generates efficient sentence-level semantic embeddings using siamese network training on Natural Language Inference (NLI) and Semantic Textual Similarity (STS) datasets.
It produces 384-dimensional dense vectors that effectively capture semantic relationships between task descriptions and object-related text.
This makes it suitable for lightweight semantic matching in edge-deployed task-aware object selection systems.

[12,13] Knowledge Distillation for Edge Deployment
Knowledge distillation traditionally transfers knowledge from large teacher models to compact student models for efficient deployment.
An offline distillation approach can instead query large language models and encode their structured semantic knowledge into compact representations.
The proposed work applies this paradigm to task-aware object selection, encoding knowledge for 14 tasks and 80 COCO categories into only 47 KB of quantised embeddings.

[14] YOLOv8-nano – Lightweight Object Detection
YOLOv8-nano provides real-time object detection across the 80 COCO categories with low computational and memory requirements.
Its CPU-only inference capability makes it suitable for edge devices and robotic platforms without dedicated GPUs.

It can therefore serve as the lightweight object detection component in an edge-deployed task-aware object selection pipeline.

METHODOLOGY:

Stage 1 - Object Detection: The input image is processed by YOLOv8-nano which detects the objects with a confidence above a minimum threshold of 0.25 along with a bounding box coordinates.

Stage 2 - Semantic Scoring: For each detected object the corresponding pre computed vector is retrieved from the knowledge bank. Cosine similarity is computed between this object embedding and the task query.

$$\text{sim}(o_i, t) = \frac{e_{o_i} \cdot e_t}{\left\| e_{o_i} \right\| \| e_t \|}$$

where $\boldsymbol{e}_{o_i}$eoi is the object embedding retrieved from the knowledge bank and $\boldsymbol{e}_t$et is the real-time encoding of the task query string.

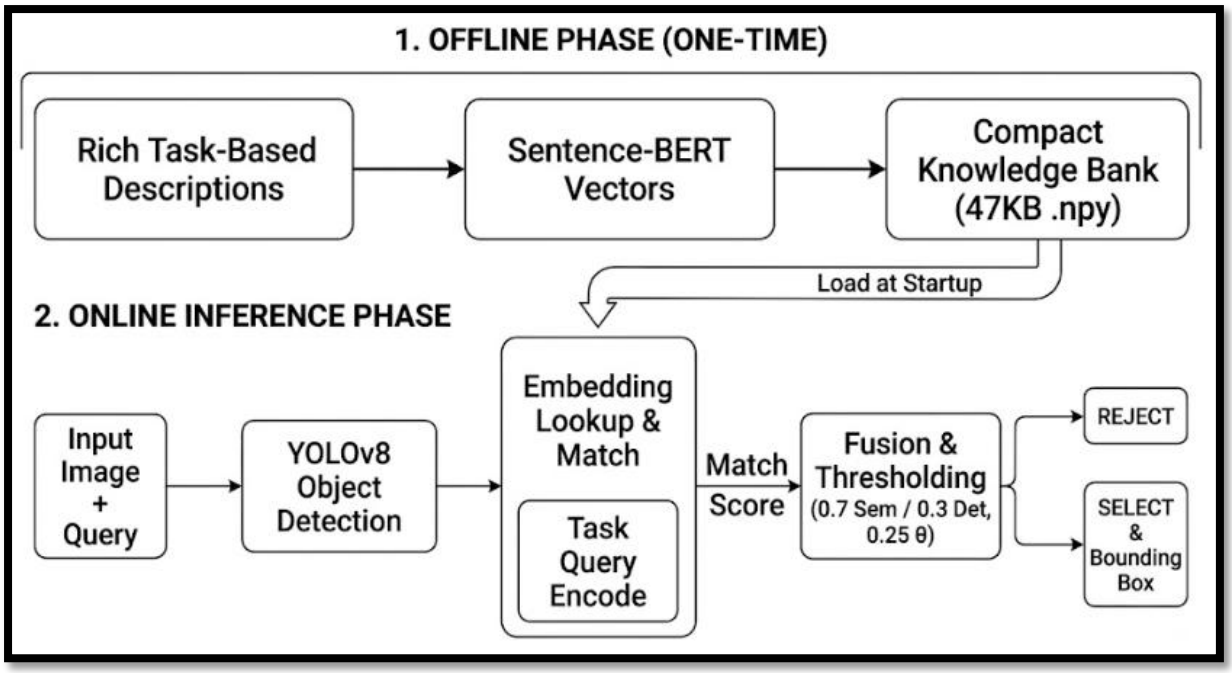
Stage 3 - Weighted Fusion: The final score for each object incorporates both semantic and detection reliability. The object score less than 0.25 are excluded from the consideration which enables the system to get the correct selection of the object.

$$score(o_i) = 0.7 \times sim(o_i, t) + 0.3 \times conf(o_i)$$

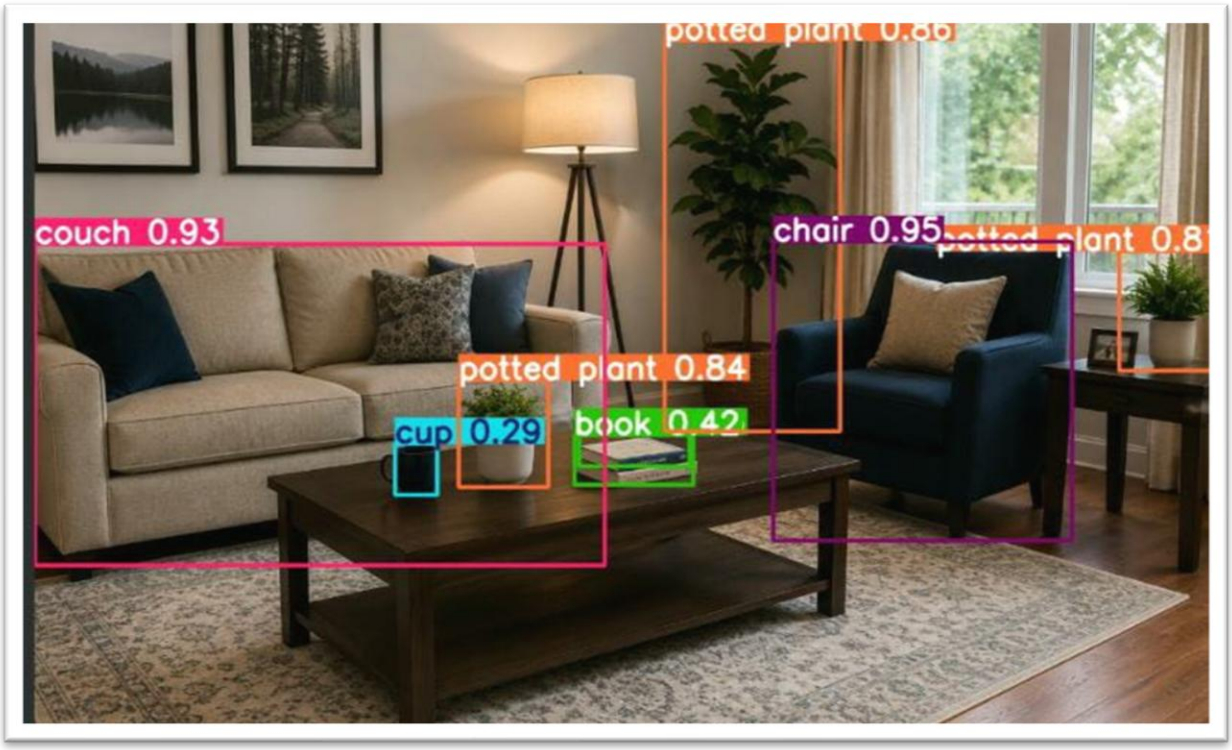
Stage 4 - Selection and Output: The highest score above the threshold will be passed as a output image

HOW OUR APPROACH DIFFERS FROM EXISTING ART:

Aspect	**Base Paper (Sawatzky et al., 2019)**	**Your Approach**
Core method	Gated Graph Neural Network (GGNN)	LLM-distilled semantic cosine similarity
Feature type	Visual features (ResNet101 CNN)	Textual semantic embeddings (Sentence-BERT)
Scene context	Graph-based object relationship modelling	Threshold-based relevance filtering
Training required	Yes — supervised on COCO-Tasks annotations	No — zero-shot, training-free
Inference dependency	Needs all objects in scene simultaneously	Each object scored independently


1. OFFLINE PHASE (ONE-TIME)
Rich Task-Based Descriptions
Sentence-BERT Vectors
Compact Knowledge Bank (47KB .npy)
Load at Startup
2. ONLINE INFERENCE PHASE
Input Image + Query
YOLOv8 Object Detection
Embedding Lookup & Match
Task Query Encode
Match Score
Fusion & Thresholding (0.7 Sem / 0.3 Det, 0.25 θ)
REJECT
SELECT & Bounding Box

OUTPUT:


couch 0.93
chair 0.95
potted plant 0.84
cup 0.29
book 0.42